WEEK 9 PROJECT REPORT

King County House Sales Data Transformation

S. Sandhya Shinde

1. Introduction

This project shows how we cleaned and changed the King County House Sales data using Microsoft Power BI. The main goal for Week 9 is to take the raw housing data and create new columns. These columns help us understand the data better and prepare it for building charts and dashboards later.

We used the Power Query Editor to look at the data and make these changes. We created new columns to calculate the age of the house when it was sold, the price for each square foot, and the total area of the property. We also grouped the houses into different categories based on their size and price. Finally, we checked all our steps and loaded the clean data into Power BI.

2. Objectives

- Import the King County House Sales data (kc_house_data.csv) into Power BI Desktop.
- Check the data to make sure all the rows and columns are correct and look healthy.
- Fix the data formats so Power BI accurately reads text fields as text and numbers as numbers.
- Create a Property Age at Sale column to find out how old the house was when it was sold.
- Create a Price per sqft column to see the cost for each square foot of living space.

- Create a Total Area column by adding the living space and the land lot size together.
- Create an Is Renovated column to show "Yes" or "No" if a house was ever remodeled.
- Group houses into a Size Bucket (Small, Medium, Large, or Luxury).
- Group houses into a Price Tier (Budget, Mid-Range, or High-End).
- Save and apply all the changes using the Close & Apply button.

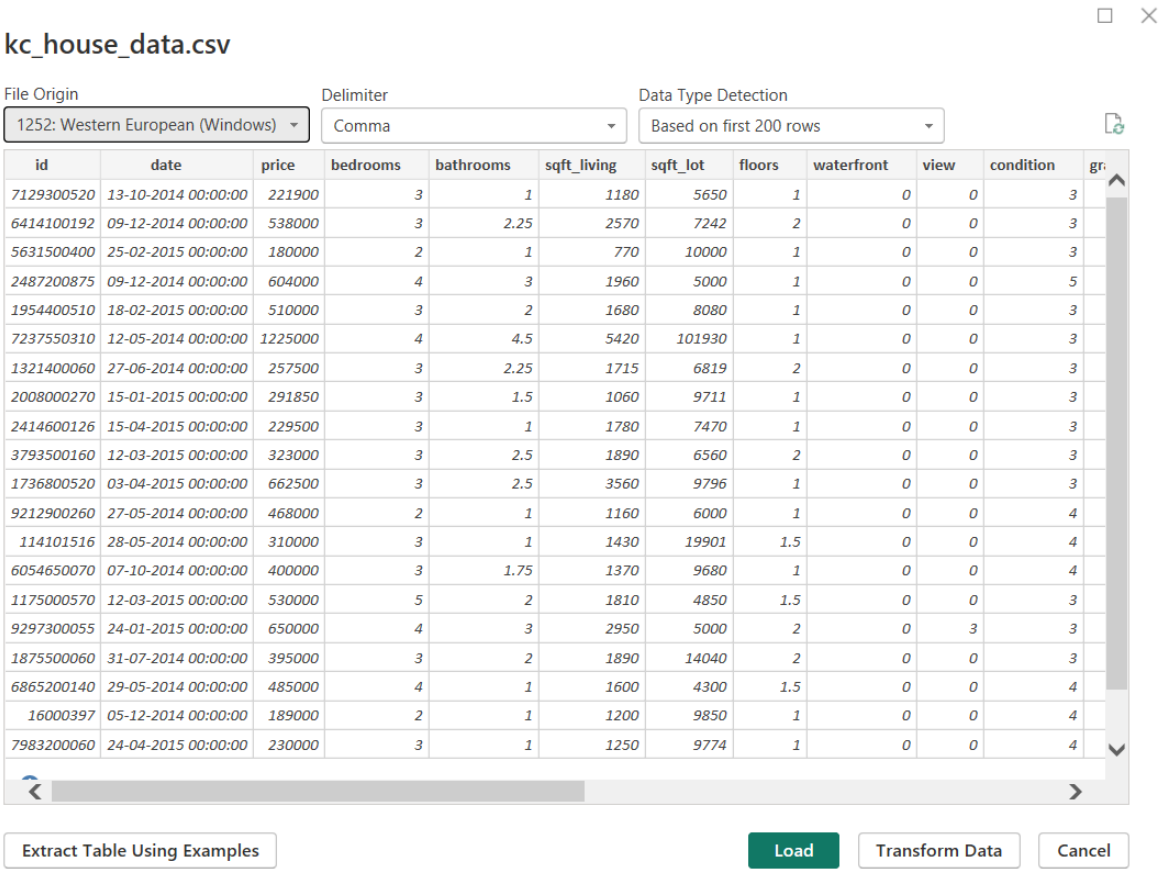
3. Tools and Technologies Used

Tool	What we used it for
Microsoft Power BI Desktop	The main software used to load data and build the final report.
Power Query Editor	The tool inside Power BI used to clean data and add new columns.
Power Query M Language	The code used by Power BI behind the scenes to run our formulas.
kc_house_data.csv	The data file containing information about house sales in Seattle.

4. Step-by-Step Data Transformation

Step 1 – Importing the Dataset

We loaded the King County House Sales data into Power BI using the Text/CSV option

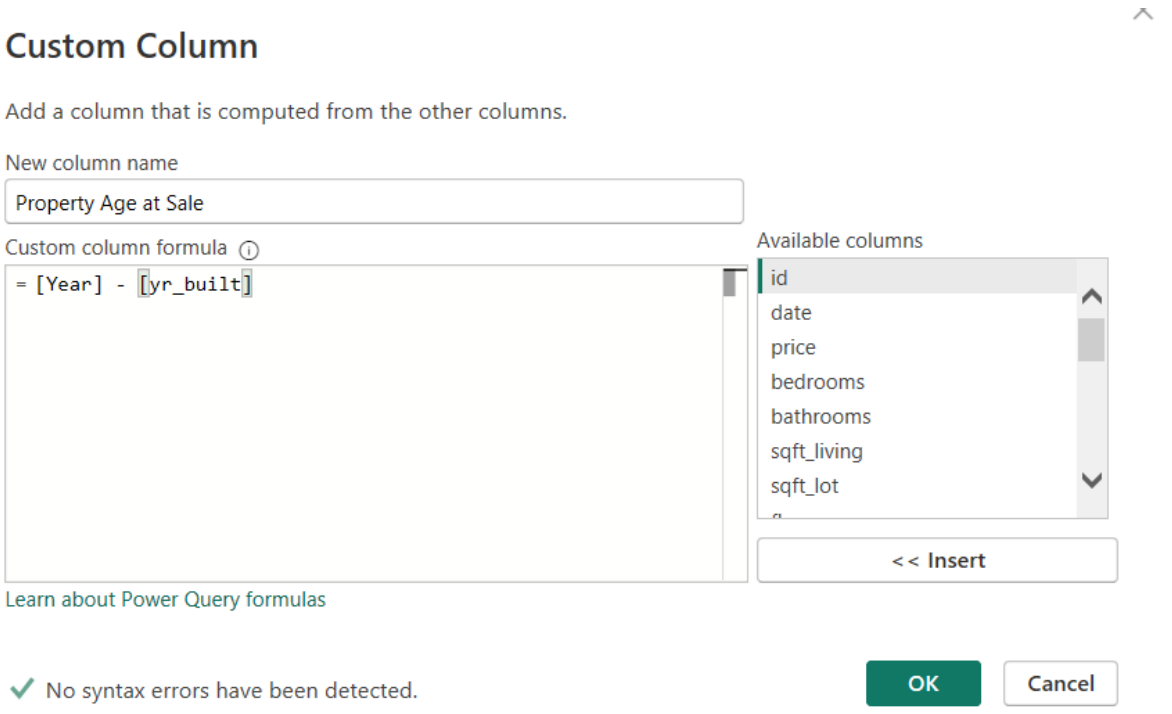


id	date	price	bedrooms	bathrooms	sqft_living	sqft_lot	floors	waterfront	view	condition	gr
7129300520	13-10-2014 00:00:00	221900	3	1	1180	5650	1	0	0	3	
6414100192	09-12-2014 00:00:00	538000	3	2.25	2570	7242	2	0	0	3	
5631500400	25-02-2015 00:00:00	180000	2	1	770	10000	1	0	0	3	
2487200875	09-12-2014 00:00:00	604000	4	3	1960	5000	1	0	0	5	
1954400510	18-02-2015 00:00:00	510000	3	2	1680	8080	1	0	0	3	
7237550310	12-05-2014 00:00:00	1225000	4	4.5	5420	101930	1	0	0	3	
1321400060	27-06-2014 00:00:00	257500	3	2.25	1715	6819	2	0	0	3	
2008000270	15-01-2015 00:00:00	291850	3	1.5	1060	9711	1	0	0	3	
2414600126	15-04-2015 00:00:00	229500	3	1	1780	7470	1	0	0	3	
3793500160	12-03-2015 00:00:00	323000	3	2.5	1890	6560	2	0	0	3	
1736800520	03-04-2015 00:00:00	662500	3	2.5	3560	9796	1	0	0	3	
9212900260	27-05-2014 00:00:00	468000	2	1	1160	6000	1	0	0	4	
114101516	28-05-2014 00:00:00	310000	3	1	1430	19901	1.5	0	0	4	
6054650070	07-10-2014 00:00:00	400000	3	1.75	1370	9680	1	0	0	4	
1175000570	12-03-2015 00:00:00	530000	5	2	1810	4850	1.5	0	0	3	
9297300055	24-01-2015 00:00:00	650000	4	3	2950	5000	2	0	3	3	
1875500060	31-07-2014 00:00:00	395000	3	2	1890	14040	2	0	0	3	
6865200140	29-05-2014 00:00:00	485000	4	1	1600	4300	1.5	0	0	4	
16000397	05-12-2014 00:00:00	189000	2	1	1200	9850	1	0	0	4	
7983200060	24-04-2015 00:00:00	230000	3	1	1250	9774	1	0	0	4	

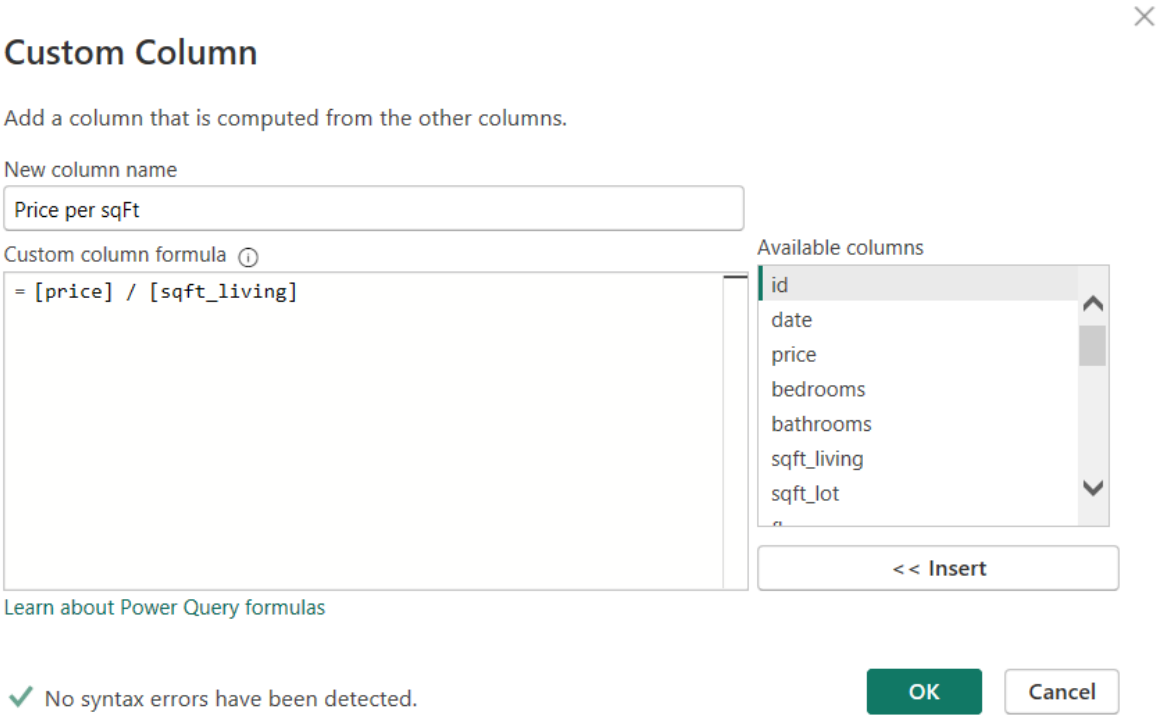
Step 2 – Creating Custom Columns

Add Column tab and click Custom Column to write formulas for four new columns:

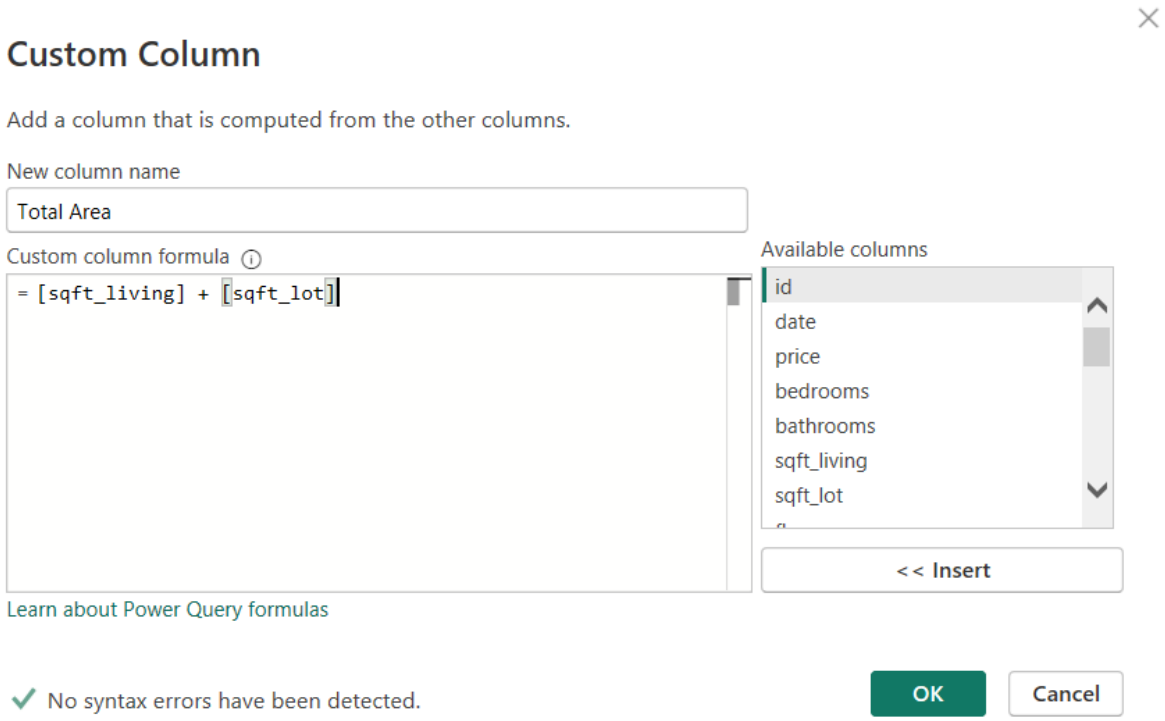
- Property Age at Sale: We used the formula = [Year] - [yr_built] to find out how old the house was on the day it was sold.



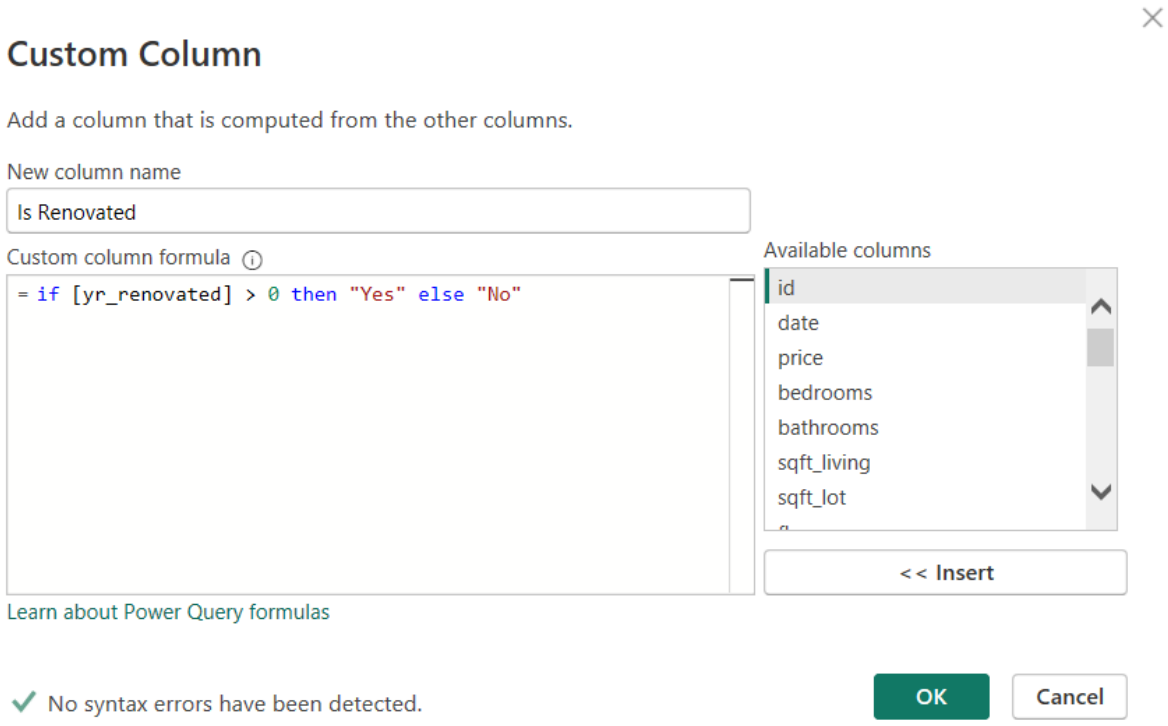
- Price per sqft: We used the formula = [price] / [sqft_living] to find the price of one square foot of living space.



- Total Area: We used the formula = [sqft_living] + [sqft_lot] to find the total size of the entire property (house + land yard).



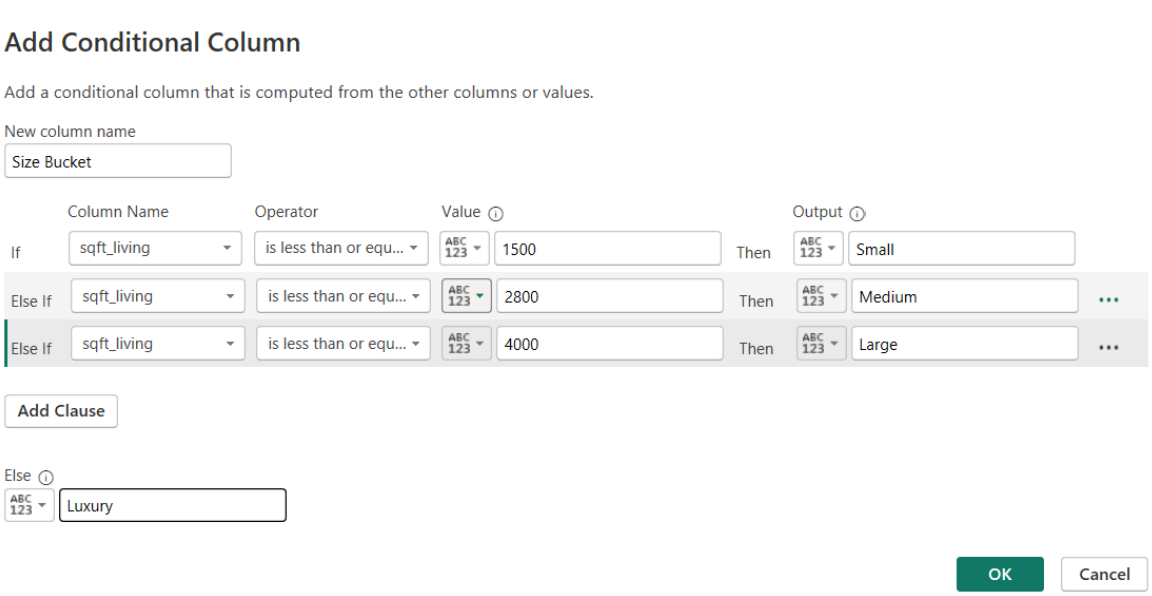
- Is Renovated: We used a simple formula = if [yr_renovated] > 0 then "Yes" else "No" to change zero values into an easy "Yes" or "No" answer.



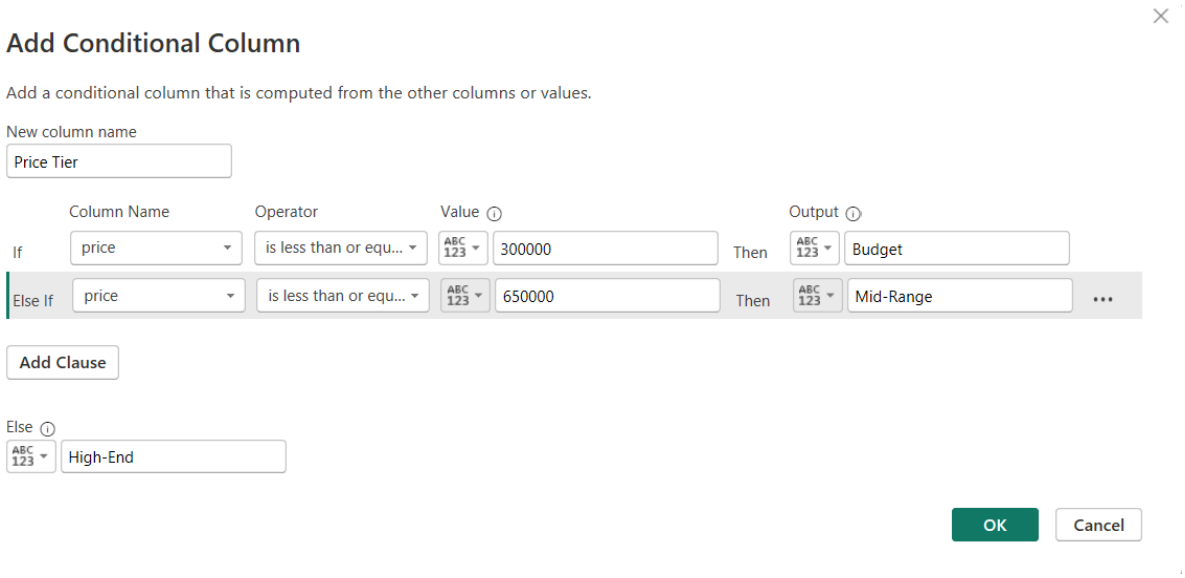
Step 3 – Creating Groups

We used the Conditional Column button to sort our data into easy-to-read groups:

- Size Bucket Column: We grouped the houses by their living space size (sqft_living) using these simple rules:
- If sqft_living is less than or equal to 1500 → Small
- Else if sqft_living is less than or equal to 2800 → Medium
- Else if sqft_living is less than or equal to 4000 → Large
- Everything else → Luxury



- Price Tier Column: We grouped the houses by their sale price using these financial rules:
- If price is less than or equal to 300000 → Budget
- Else if price is less than or equal to 650000 → Mid-Range
- Everything else → High-End



5. Conclusion

In Week 9, we successfully cleaned and improved the King County House Sales data using Power Query. By creating new custom columns, we can now easily find out the age of a house when it was sold, see its price per square foot, and check if it was ever remodeled.

Grouping the houses into different size and price categories will make filtering the information much simpler. The dataset is now clean, and well-organized.